The oscillator is used in research and measurement of celestrial navigation, space tracking purpose, and the timing signal in the measuring instruments and medical devices. There is variety of industrial applications of crystal oscillator such as computers, digital systems, marine, modems, sensors, telecommunications and disk drives. It is used in automotive field by engine controlling, stereo and in GPS system. It is also used in consumer applications like TV systems, PCs, video games, toys, radio systems and cellular phones.

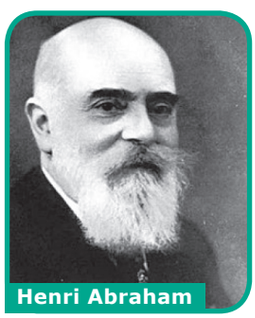


Which oscillator circuit used for medical treatment?

Multi wave oscillator is applied for the treatment of cancer.

7.6. MULTIVIBRATORS

Multivibrators are two stage switching circuits in which the output of the first stage is fed to the input of the second stage and vice versa. The outputs of the two stages are complementary (high or low). A specific characteristic of multivibrator is that it uses passive elements like resistor and capacitor to determine the output state.


Henri Abraham

History: The first multivibrator circuit, the astable multivibrator oscillator was invented by Henri Abraham and Eugene Bloch in 1920. They called their circuit 'a' multivibrator because its output waveform was rich in harmonics.

Multivibrators are of three types namely

1. Astable multivibrator
2. Monostable multivibrator
3. Bistable multivibrator

7.6.1. Astable Multivibrator

The astable or free running multivibrator generates square wave without any external triggering pulse. It has no stable states, i.e. it has two quasi-stable states. It switches back and forth from one state to the other, remaining in each state for a time depending upon the discharging of a capacitive circuit.

7.6.2. Monostable Multivibrator

Monostable multivibrator has one stable and one quasi-stable. It is also known as one- shot multivibrator. In this, the output pulse duration is determined by the RC time constant and is given as *1.11 * R * C*. This multivibrator cannot generate square waves of its own like an astable multivibrator. One external triggering pulse will cause it to generate the rectangular waves.

7.6.3. Bistable Multivibrator



The bistable multivibrator has two stable states. A triggering pulse is applied to the circuit causes it to switch from one state to other. Another trigger pulse is then required to switch the circuit back to its original state.

Construction: Figure 7.9 shows the circuit of a bistable multivibrator using two NPN transistors. In this circuit, the output of a transistor Q_2 is coupled to the base of

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